

Experiment 2

Sumit Roy

23MS090

Group: A02

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1 Title

Study of Characteristics of n-p-n BJT

2 Aim

To study the input and output characteristics of an NPN Bipolar Junction Transistor (BJT) in the Common Emitter (CE) configuration and to observe its input and output characteristics and determine current gain (β).

3 Theory

3.1 Bipolar Junction Transistor

A Bipolar Junction Transistor (BJT) is a three-terminal semiconductor device consisting of two p-n junctions. It has three terminals: Emitter (E), Base (B), and Collector (C). In an NPN transistor, a thin, lightly doped p-type base is sandwiched between two n-type regions (emitter and collector). The emitter is highly doped.

For the transistor to operate in the active region (used for amplification):

- The Emitter-Base junction is forward-biased.
- The Collector-Base junction is reverse-biased.

So, the first junction faces almost negligible resistance where the second junction faces very high resistance while the current flowing through them remains almost same (very low recombination via base) and it feels like the resistance got transferred. that's how the device got its name: transfer + resistor = transistor.

The relation of currents through each part is:

$$I_E = I_B + I_C$$

As collector current is mainly the non recombined part from emitter current mixed with reverse saturation current (I_{CBO}) we can write:

$$I_C = \alpha I_E + I_{CBO}$$

From those equations we get:

$$I_C = \frac{\alpha}{1 - \alpha} I_E + \frac{1}{1 - \alpha} I_{CBO}$$

Taking $\beta = \frac{\alpha}{1 - \alpha}$ we get:

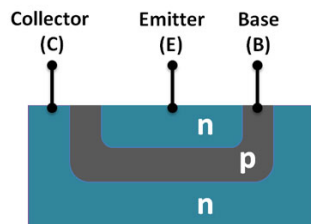
$$I_C = \beta I_B + (\beta + 1) I_{CBO}$$

Now, the reverse saturation current is very small hence negligible:

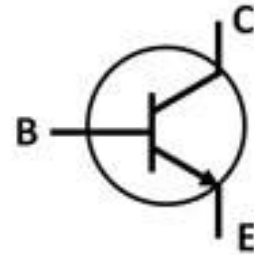
$$I_C = \beta I_B$$



(a) 3D image



(b) Inner Structure



(c) Circuit representation

Figure 1: n-p-n Bipolar Junction Transistor

3.2 Common Emitter Configuration

In this configuration, the emitter terminal is common to both the input and output circuits.

- **Input parameters:** Base Current (I_B) and Base-Emitter Voltage (V_{BE}).
- **Output parameters:** Collector Current (I_C) and Collector-Emitter Voltage (V_{CE}).

3.2.1 Input Characteristics

The input characteristic is the curve plotted between input current (I_B) and input voltage (V_{BE}) at a constant output voltage (V_{CE}). It resembles the characteristic of a forward-biased p-n junction diode because after hitting the threshold voltage (V_{BE}) the majority carriers start flowing. Also as we increase V_{CE} , it also increases V_{CB} which causes less recombination at base (early effect) which decreases the base current I_B .

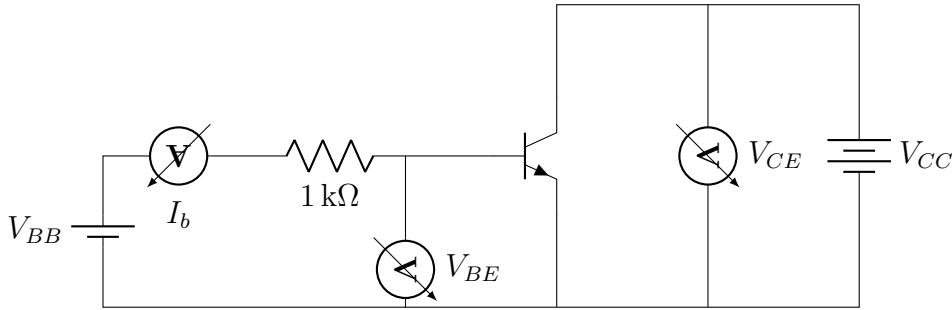


Figure 2: Circuit diagram for observing input characteristics

3.2.2 Output Characteristics

The output characteristic is the curve plotted between output current (I_C) and output voltage (V_{CE}) at a constant input current (I_B). The transistor operates in three regions: Cut-off, Active, and Saturation. The ratio of collector current to the base current gives the current amplification factor (β) (as stated earlier):

$$\beta = \left(\frac{I_C}{I_B} \right)_{V_{CE}=\text{constant}} \quad (1)$$

- **Saturation Region:** In the saturation region, both the emitter–base and collector–base junctions are forward biased beyond the cut-in voltage (V_γ). Since the forward-bias voltages V_{EB} and V_{CB} are each only a few tenths of a volt, the resulting collector–emitter saturation voltage, given by

$$V_{CE(\text{sat})} = V_{CB} - V_{EB},$$

is also typically only a few tenths of a volt.

Under saturation, the transistor can no longer maintain normal current amplification. For a fixed collector supply and load, the collector current is mainly determined by the external circuit and shows very little dependence on the base current.

- **Active Region:** In the active region, the emitter junction is forward biased while the collector junction remains reverse biased. The collector-emitter voltage is greater than the saturation voltage, which is usually around 0.1–0.2 V for low-power transistors.

Within this region, the output characteristics are nearly straight and evenly spaced, indicating proper amplification. For low-distortion amplifier operation, the transistor should be kept in this region. So, we will measure β in this region.

- **Cut-off Region:** When $I_B = 0$, Then $I_C = (\beta + 1)I_{CBO}$. Now, I_{CBO} maybe negligible but β has a big value. So The Collector current will still flow even if input (base current) is turned off.

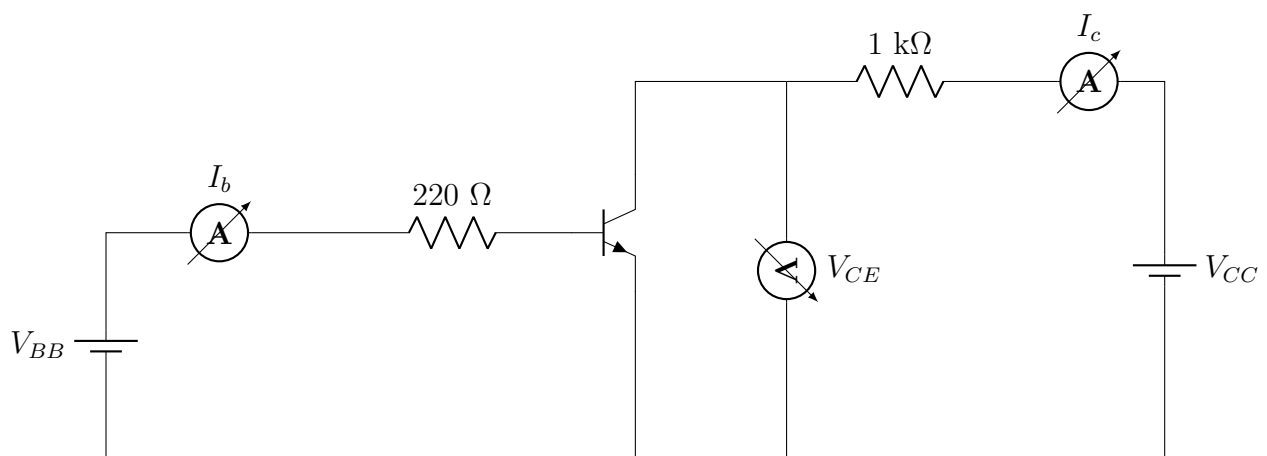


Figure 3: Circuit diagram for output characteristics

4 Data

Table 1: Experimental Observations for Input Characteristics

Sl. No	$V_{CE} = 2 \text{ V}$		$V_{CE} = 3 \text{ V}$		$V_{CE} = 4 \text{ V}$	
	$V_{BE} \text{ (V)}$	$I_B \text{ (}\mu\text{A)}$	$V_{BE} \text{ (V)}$	$I_B \text{ (}\mu\text{A)}$	$V_{BE} \text{ (V)}$	$I_B \text{ (}\mu\text{A)}$
1.	0.00	0	0.00	0	0.00	0
2.	0.24	0	0.28	0	0.31	0
3.	0.41	0	0.44	0	0.46	0
4.	0.51	1	0.53	1	0.54	1
5.	0.54	2	0.56	2	0.57	2
6.	0.57	6	0.58	5	0.59	4
7.	0.60	14	0.61	11	0.61	9
8.	0.62	32	0.63	28	0.64	24
9.	0.64	65	0.65	58	0.66	52
10.	0.66	115	0.67	108	0.67	98
11.	0.67	180	0.68	165	0.68	148
12.	0.68	255	0.69	235	0.69	215
13.	0.69	445	0.70	415	0.71	390

Table 2: Experimental Observations for Output Characteristics

Sl. No	$I_B = 10 \mu A$		$I_B = 20 \mu A$		$I_B = 30 \mu A$		$I_B = 40 \mu A$	
	V_{CE} (V)	I_C (μA)	V_{CE} (V)	I_C (μA)	V_{CE} (V)	I_C (μA)	V_{CE} (V)	I_C (μA)
1.	0.03	77	0.016	87	0.011	78	0.009	79
2.	0.04	165	0.024	178	0.018	183	0.015	186
3.	0.05	259	0.032	275	0.024	276	0.019	279
4.	0.06	353	0.037	362	0.029	376	0.024	381
5.	0.062	433	0.041	458	0.033	467	0.027	472
6.	0.066	532	0.046	549	0.036	560	0.030	567
7.	0.076	612	0.056	717	0.043	732	0.035	745
8.	0.079	705	0.061	933	0.050	946	0.041	948
9.	0.082	818	0.066	1099	0.053	1113	0.045	1118
10.	0.085	885	0.075	1380	0.060	1393	0.051	1403
11.	0.098	1061	0.083	1657	0.066	1674	0.056	1682
12.	0.113	1337	0.090	1930	0.072	1949	0.061	1956
13.	0.135	1601	0.108	2490	0.082	2510	0.070	2520
14.	0.292	1782	1.31	2990	0.091	3030	0.078	3040
15.	0.738	1730	1.454	3740	0.15	5030	0.12	5070
16.	1.334	1760	4.17	4160	2.66	5660	0.69	7580
17.	3.30	1850	8.67	3790	6.61	5880	4.65	7820
18.	6.40	1890	—	—	—	—	—	—
19.	10.54	1940	—	—	—	—	—	—

5 Analysis

5.1 Analyzing Input Characteristics

The input characteristic curves (I_B vs. V_{BE}) for $V_{CE} = 2V, 3V$, and $4V$ were plotted. The graph shows that for small values of base-emitter voltage, the base current is negligible. Once V_{BE} exceeds the knee voltage ($\approx 0.6V$), I_B increases exponentially. We also noticed as we decrease V_{CE} , the rate of base current increment goes higher, supporting its theory.

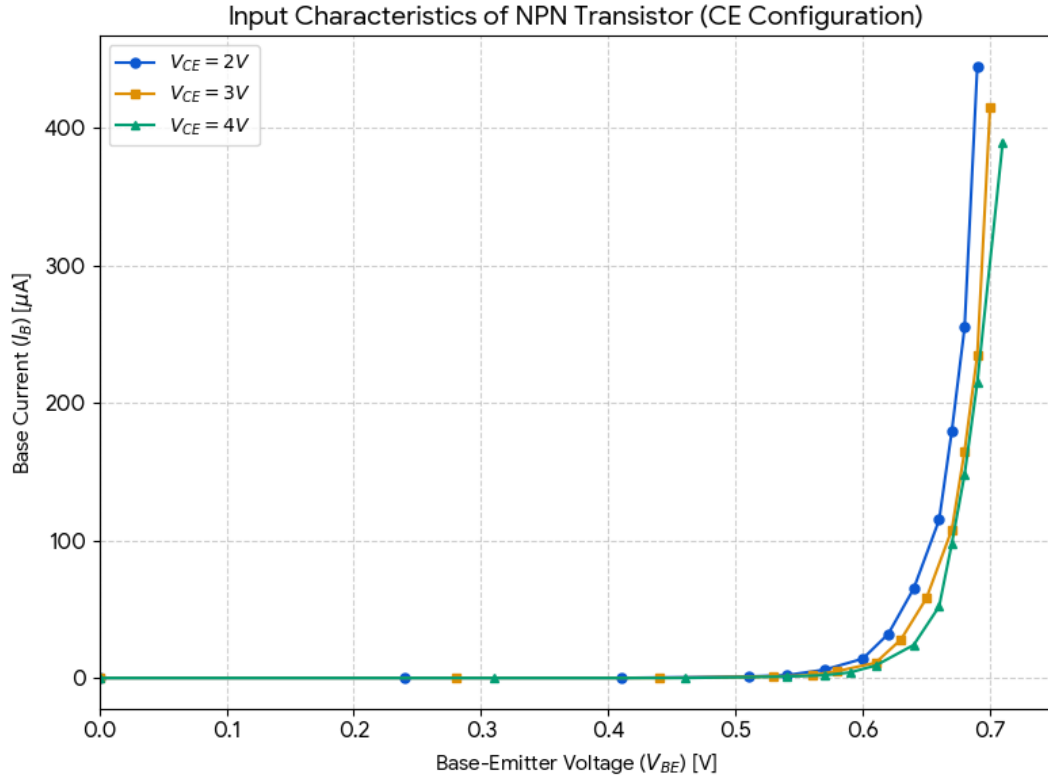


Figure 4: Plot of Input Characteristics

5.2 Analyzing Output Characteristics

The output characteristic curves (I_C vs. V_{CE}) were plotted for fixed base currents of $10\mu A$, $20\mu A$, $30\mu A$, and $40\mu A$. The graph displays the active region where I_C is nearly independent of V_{CE} and primarily determined by I_B .

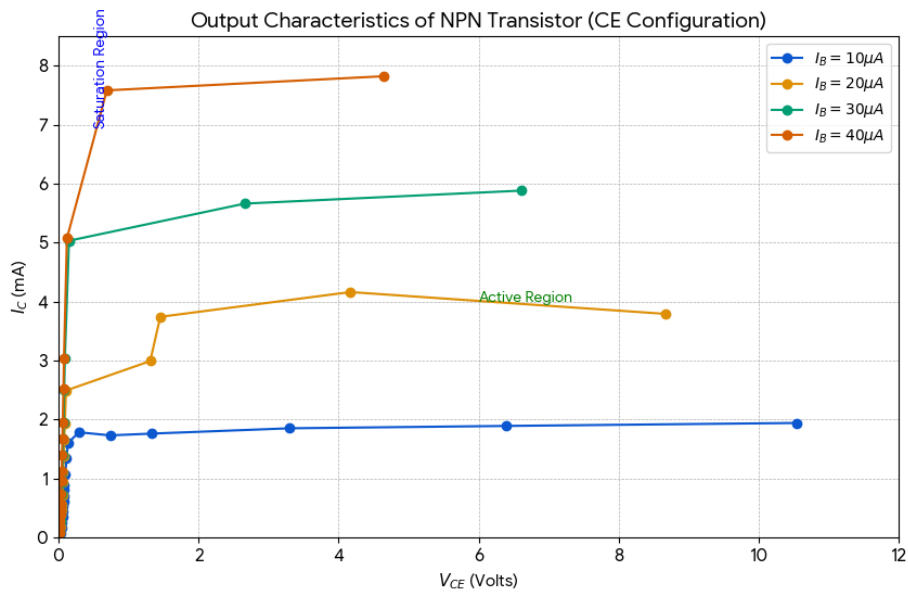


Figure 5: Plot of Output Characteristics

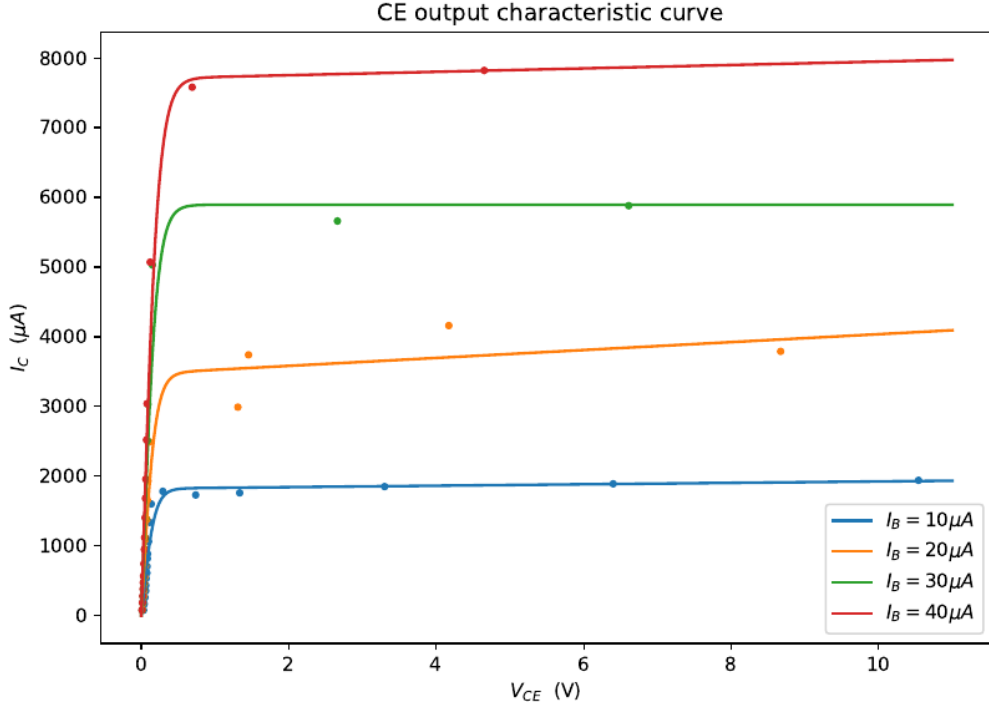


Figure 6: Fitted Plot of Output Characteristics

5.2.1 Calculation of DC Current Gain (β)

The DC current gain β is defined as the ratio of the collector current (I_C) to the base current (I_B) at a constant collector-emitter voltage (V_{CE}).

$$\beta = \frac{I_C}{I_B} \quad (2)$$

- **Case 1: At $V_{CE} = 5$ V** Using the experimental data interpolated at $V_{CE} = 5$ V. As in our data the maximum recorded V_{CE} for $I_B = 40 \mu A$, was less than 5 V, it is excluded from this calculation.

Table 3: Calculation of Current Gain (β) at $V_{CE} = 5$ V

Sl. No.	I_B (μA)	I_C at 5V (μA)	$\beta = \frac{I_C}{I_B}$
1.	10	1872	187.2
2.	20	4092	204.6
3.	30	5790	193.0

Average Current Gain ($V_{CE} = 5$ V):

$$\beta_{avg(5V)} = \frac{187.2 + 204.6 + 193.0}{3} \approx \mathbf{194.9}$$

- **Case 2: At $V_{CE} = 4$ V** Using the experimental data interpolated at $V_{CE} = 4$ V. In this case, valid data points exist for all base currents.

Table 4: Calculation of Current Gain (β) at $V_{CE} = 4V$

Sl. No.	I_B (μA)	I_C at 4V (μA)	$\beta = \frac{I_C}{I_B}$
1.	10	1859	185.9
2.	20	4134	206.7
3.	30	5735	191.2
4.	40	7781	194.5

Average Current Gain ($V_{CE} = 4V$):

$$\beta_{avg(4V)} = \frac{185.9 + 206.7 + 191.2 + 194.5}{4} \approx \mathbf{194.6}$$

6 Sources of error

1. **Heating Effect:** The transistor may heat up during operation, causing fluctuations in readings, particularly in the saturation region.
2. **Instrument Precision:** The limited resolution of the multimeter could introduce reading errors, especially for small base currents.
3. **Connection Resistance:** Loose connections on the breadboard may add stray resistance to the circuit.

7 Conclusion

The input and output characteristics of the NPN Bipolar Junction Transistor in the Common Emitter configuration were successfully studied.

The input characteristics curve resembles that of a forward-biased diode, with a cut-in voltage of approximately 0.6 V. It was observed that the base current (I_B) increases slightly as the collector-emitter voltage (V_{CE}) decreases for a fixed base-emitter voltage (V_{BE}), consistent with the Early Effect.

The output characteristics demonstrated three distinct regions: cut-off, active, and saturation. In the active region, the collector current (I_C) was found to be nearly independent of V_{CE} and controlled primarily by I_B . The DC current gain (β) was calculated from the active region data, yielding an average value of $\beta \approx 195$. The consistency of β across different operating points confirms the linear amplification capability of the transistor within the active region.